Ray Diagrams

Exam code: 0625 & 0972

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Features of ray diagrams

- Ray diagrams can be described using the following terms:
  - Principal axis
  - Principal focus, or focal point
  - Focal length
- The principal axis is defined as:

A line which passes through the centre of a lens

- The principal focus, or focal point, is defined as:

The point at which rays of light travelling parallel to the principal axis intersect the principal axis and converge or the point at which diverging rays appear to proceed

- Focal length is defined as:

The distance between the centre of the lens and the principal focus

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

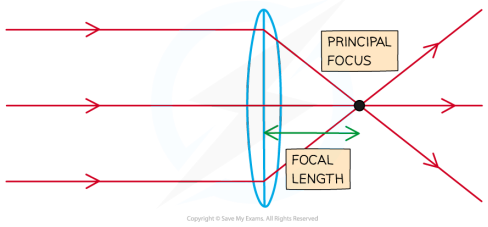
Converging & Diverging Lenses

- A lens is a piece of equipment that forms an image by **refracting** light

Converging lenses

- In a converging lens, parallel rays of light are brought to a focus
  - This point is called the **principal focus**
- This lens is sometimes referred to as a **convex** lens
- The distance from the lens to the principal focus is called the **focal length**
  - This depends on how **curved** the lens is
  - The **more** curved the lens, the **shorter** the focal length

Converging lens diagram

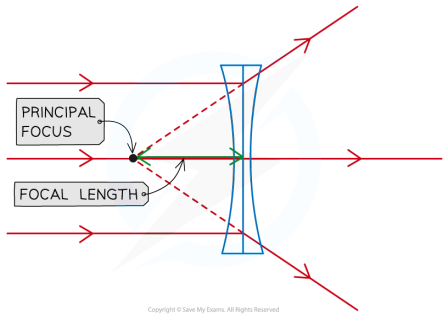


The focal length is the distance from the lens to the principal focus

Diverging lenses

- In a diverging lens, parallel rays of light are made to diverge (spread out) from a point
  - This lens is sometimes referred to as a **concave** lens
- The principal focus is now the point from which the rays appear to diverge from

Diverging lens diagram

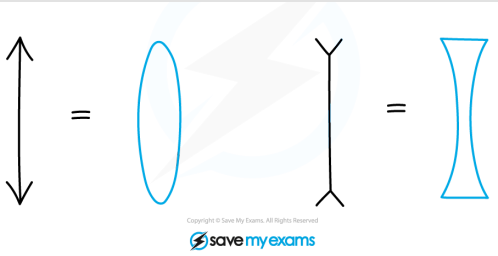


Parallel rays from a diverging lens appear to come from the principal focus

Representing lenses

- In diagrams, the following symbols are often used to represent each type of lens:

Converging and diverging lens symbols



Concave and convex lens symbols

Examiner Tips and Tricks

To remember which lens is converging or diverging, think of the following: Con**v**ex lens = Con**v**erging. You need to be able to describe how the lenses affect the light rays.

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